

RENDY RAMADHANA WAHYUDI

PHYSICS INSTRUMENTATION | OILFIELD ENTHUSIAST

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PROFESSIONAL SUMMARY

Physics Engineer with strong field experience in advanced sensor systems and instrumentation, eager to contribute to the evolving energy landscape. Dedicated to integrating hardware-software solutions for efficient and reliable operational outcomes. Committed to safety-first principles, operational excellence, and team coordination in dynamic environments, seeking to drive innovation within an integrated energy company.

CORE COMPETENCIES

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|---------------------------------------|--|
| ✓ Control System Expertise | ✓ Instrumentation & Measurement Design |
| ✓ System Integration & Validation | ✓ Brownfield Project Management |
| ✓ Equipment Reliability & Maintenance | ✓ Process Automation |
| ✓ Safety & Fire/Gas Systems | ✓ Digital Security & Critical Comms |

LEADERSHIP EXPERIENCE

Program Coordinator, Village Empowerment, HiMA GEOFISIKA PEDRA, UNPAD | Sep 2021 - Feb 2022

Coordinated long-term rural outreach, trained disaster awareness to 100+ residents.

- Conducted social mapping across 50+ households and collaborated to co-design impactful educational agendas.
- Delivered earthquake preparedness workshops attended by 100+ residents, enhancing local resilience.

Vice Chairman, POA School of Leader XIII UNPAD | Oct 2020 - Nov 2020

Managed a climate-focused social project, achieving target outcomes through efficient collaboration.

- Managed and oversaw the performance of 7 divisions utilizing tools such as Work Breakdown Structure (WBS), Trello, and Google Calendar for project management and scheduling.
- Applied fundamental management principles and business acumen to optimize resources and achieve activity targets, successfully attaining an 80 % project success indicator.

PROFESSIONAL EXPERIENCE

Barista (Fixed-Term Contract), PT. Mahat Masagi UNPAD | Aug 2024 - Present

Applied hospitality principles in cafe operations to ensure customer satisfaction and service efficiency.

- Managed shift schedules, master recipes, and raw material inventory, and oversaw operational costs using spreadsheets. Successfully increased engagement and upselling, resulting in a 2x revenue increase.

Instrumentation Engineer (Contract-Based), Geophysics UNPAD Research Team | Jul 2023 - Aug 2023

Developed systems under tight deadlines, applying various approaches to enhance operational effectiveness.

- Initiated Instrumentation performance optimization using open-source numerical simulation software, resulting in up to 50% savings in cost and material use.
- Designed and built a functional prototype that served as the foundation for measurable and sustainable further research development.

EDUCATION

Master of Science in Physics Engineering, Universitas Padjadjaran, West Java | Nov 2023 - Aug 2025 (Expected)

- **Capstone Project:** *Study and Implementation of a Distributed Wireless IoT-Based Ground Vibration Recording System for Environmental Monitoring.*

Bachelor of Science in Geophysical Engineering, Universitas Padjadjaran, West Java | Aug 2019 - Jul 2023

- **Capstone Project:** *Prototype development of IoT-based distributed seismic sensor network*, published at DOI: 10.2991/978-2-38476-283-5_7.

CERTIFICATIONS

Basic Electrical Safety for the O&G Industry, Belajar Elektro | Jun 2025

IWCF Level 1, SkillsVX | May 2025

Well Operations Crew Resource Management (WOCRM), SkillsVX | Apr 2025

TOEFL ITP, Universitas Indonesia | Jul 2024

TECHNICAL SKILLS

Programming & Databases: Ladder Logic Diagram, Sequential Function Chart, Function Block Diagram, Instruction List, Structured Text, Python, C++, MATLAB, SQL | **Instrumentation & Embedded Systems:** PLC, Arduino, STM32, ESP32, Keil uVision | **Industrial Communication Protocols:** MODBUS (RTU, TCP/IP), HART, PROFIBUS (DP, PA), PROFINET, EtherNet/IP, OPC UA | **Engineering & Simulation Software:** LabVIEW, InstruCalc, Automation Studio, ETAP, PSCAD, PowerWorld Simulator, PSpice, NI Multisim, OrCAD, Simulink | **Design & CAD:** AutoCAD Electrical, SolidWorks, Eagle, Revit | **ERP & Office Software:** SAP (ERP System), Microsoft Excel, Microsoft Word